Engineering Department Doc, General Developer 1 Final Handoff Summary

Final update after Layer 2 v0.2.3 Shape Metric Quality Flags and Layer 3 Readiness Filter

Document purpose: This is the final closeout summary for General Developer 1. It summarizes accepted Layer 1 and Layer 2 implementation work and records the final Layer 2 v0.2.3 readiness filter. Future developers should use this as a navigation document, not as a replacement for the detailed engineering handoff documents.

1. Documents Future Developers Must Read

Document	Purpose
AIB Project System Architecture / Workflow Specification	Explains the full layered system, project philosophy, AI boundaries, and how each layer connects to the broader Soccer TIPS workflow.
Engineering Department Doc, Validation & Data Ingestion Handoff Summary	Explains why Layer 1 ingestion rules exist, including SkillCorner and Metrica dataset behavior, parser rules, coordinate handling, missing-file behavior, and validation findings.
Engineering Department Doc, Layer 1_v0.1 General Developer Handoff Notes	Explains reusable Layer 1 ingestion, canonical processed inputs, batch processing, QA, and expected warnings.
Engineering Department Doc, Layer 2_v0.2 General Developer Handoff Notes	Explains Layer 2 feature generation, row grains, QA, football sanity validation, and scope boundaries.
Engineering Department Doc, Layer 2 v0.2.1 team_width Warning Review Result	Explains the team_width warning review, the semantic mismatch discovery, and why the v0.2.2 patch was required.
Engineering Department Doc, Layer 2 v0.2.2 team_width-team_depth Patch Result	Explains the metric patch, revalidation, residual strict-width warnings, and post-patch football sanity result.
Engineering Department Doc, Layer 2 v0.2.3 Shape Metric Quality Flags and Layer 3 Readiness Filter	Explains the final non-destructive quality flag table and how Layer 3 should consume shape metrics safely.

2. Work Completed by General Developer 1

Layer 1 work

- Built provider-specific ingestion loaders for SkillCorner and Metrica.
- Created shared canonical player and ball tracking outputs for downstream use.
- Preserved raw provider coordinates and generated metric coordinates separately where needed.
- Implemented validation report creation and controlled Layer 1 batch processing.
- Added Layer 1 QA inspection with schema compliance checks.
- Supported SkillCorner Open Data and Metrica Sample_Game_1 / Sample_Game_2; Metrica Sample_Game_3 remains intentionally out of scope.

Layer 2 work

- Created the src/soccer_tips/features/ package.
- Implemented SkillCorner phase-context linking using phases_of_play.csv.
- Computed frame-level team shape metrics from Layer 1 canonical tracking outputs.
- Aggregated frame-level metrics into phase-segment summaries and match tactical profiles.
- Built Layer 2 tests, build runner, QA utility, and batch support across all processed SkillCorner matches.

- Built Layer 2 v0.2 football sanity validation outputs, including phase metric summaries, structured JSON summary, and validation plots.
- Reviewed the team_width pitch-bound warning, identified and patched a team_width/team_depth semantic mismatch, and reran validation.
- Built Layer 2 v0.2.3 shape metric quality flags and Layer 3 readiness filter outputs.

3. Final Accepted Project State

- Layer 1 v0.1 is accepted as reusable ingestion infrastructure.
- Layer 2 v0.1 is accepted as the feature-engineering foundation.
- Layer 2 v0.2 football sanity validation is completed and now returns status pass after the width/depth patch.
- Layer 2 v0.2.1 found a team_width/team_depth semantic mismatch.
- Layer 2 v0.2.2 patched the semantic mismatch and reran tests, Layer 2 batch, Layer 2 QA, football sanity QA, and team_width review.
- Layer 2 v0.2.3 added non-destructive shape metric quality flags and a Layer 3 readiness filter.
- Shape quality QA status is pass: 873,296 rows checked, 873,296 included for Layer 3, 0 excluded, 0 tolerance failures, 0 low-player-count exclusions.
- SkillCorner remains the primary provider for phase-aware Layer 2 work because it has validated phases_of_play data.
- Metrica is supported for Layer 1 ingestion but is not currently used for phase-aware Layer 2 analysis.
- No Layer 3 modeling, clustering, tactical identity modeling, AI interpretation, dashboards, reports, or recommendation systems have been added yet.

4. Current Code Structure

Folder / file area	Use
src/soccer_tips/data/	Layer 1 ingestion, provider-specific loaders, coordinate normalization, validation utilities, Layer 1 build runner, and Layer 1 QA inspection.
src/soccer_tips/features/	Layer 2 schemas, phase-context linking, orientation placeholder module, patched team shape metrics, aggregation, Layer 2 build runner, Layer 2 QA, football sanity validation, validation plotting, team_width review utility, and shape quality flags.
tests/	Lightweight tests for Layer 1 ingestion, Layer 2 feature engineering, football sanity helpers, and shape metric quality flags.
data/processed/	Local generated outputs from Layer 1 and Layer 2. This is reproducible output data, not manually edited source material.
data/processed/features/	Layer 2 feature outputs, QA summaries, football sanity summaries, team_width review outputs, and layer2_shape_metric_quality_flags.csv.
outputs/figures/layer2_sanity/	Validation-only Layer 2 football sanity figures and team_width review plots. These are sanity plots, not dashboards or final reports.

5. Important Run Commands

Task	PowerShell command
Activate virtual environment	.\venv\Scripts\Activate
Install requirements	pip install -r requirements.txt
Set PYTHONPATH	$env:PYTHONPATH="src"

Run Layer 1 tests	python tests/test_layer1_ingestion.py
Run Layer 1 batch	python -m soccer_tips.data.build_layer1 --provider all --batch
Run Layer 1 QA	python -m soccer_tips.data.qa_layer1_outputs
Run Layer 2 tests	python tests/test_layer2_features.py
Run football sanity tests	python tests/test_football_sanity.py
Run shape quality tests	python tests/test_shape_quality.py
Run Layer 2 batch	python -m soccer_tips.features.build_layer2 --provider skillcorner --batch
Run Layer 2 QA	python -m soccer_tips.features.qa_layer2_outputs
Run Layer 2 football sanity validation	python -m soccer_tips.features.qa_football_sanity
Run shape quality QA	python -m soccer_tips.features.qa_shape_quality

6. Key Technical Decisions Made

- Use provider-specific loading logic with shared canonical outputs.
- Use SkillCorner phases as the trusted phase-context layer for current phase-aware Layer 2 work.
- Preserve raw provider coordinates and metric coordinates separately.
- Keep orientation normalization intentionally inactive until it is validated and approved.
- Preserve detection and extrapolation flags for quality diagnostics and future confidence filtering.
- Treat missing SkillCorner dynamic events as a warning, not a failure.
- Keep Metrica Sample_Game_3 out of current scope because it uses a different internal format.
- Start Layer 2 with simple, interpretable team shape metrics before deeper tactical interpretation.
- Patch team_width/team_depth to the documented convention: team_width = y_metric range and team_depth = x_metric range.
- Use non-destructive shape quality flags instead of deleting rows or changing canonical Layer 2 schemas.
- Do not add Layer 3 modeling or AI interpretation before descriptive tactical profiles exist.

7. Known Warnings and Limitations

- SkillCorner match 1953632 is missing optional dynamic_events.csv. This is expected and non-fatal.
- Metrica Sample_Game_1 contains one known kickoff initialization event range issue. This is expected and non-fatal.
- Orientation normalization is inactive. Current analysis uses Layer 1 metric coordinates.
- The previous team_width/team_depth semantic mismatch was found, patched, and revalidated.
- After the patch, football sanity QA status is pass.
- Residual strict team_width pitch-bound warnings remain: 256 rows, but 0 width tolerance failures.
- Shape quality QA includes all rows for Layer 3 under the current include rule, but high extrapolated-count/rate warnings affect 258,068 rows. These are caution flags, not automatic exclusions.
- centroid_x should not be used for strong directional claims until attacking direction/orientation is validated.
- There are no final tactical identity claims yet.
- There is no clustering yet.
- There is no AI interpretation yet.
- There is no dashboard or reporting layer yet.
- Current Layer 2 outputs are feature and validation outputs, not final scouting conclusions.

8. What Future Developers Should NOT Redo

- Do not redo Layer 1 ingestion from scratch unless a validated defect requires it.

- Do not change canonical schemas casually.
- Do not rebuild SkillCorner or Metrica loaders without a clear reason.
- Do not treat raw tracking possession as authoritative without validation.
- Do not activate orientation normalization casually.
- Do not reverse or rename team_width/team_depth. The patched convention is team_width = y_metric range and team_depth = x_metric range.
- Do not jump straight to clustering.
- Do not use Layer 2 metrics for final tactical claims without sanity validation, quality flags, and Lead Developer review.
- Do not use shape metrics in Layer 3 without joining layer2_shape_metric_quality_flags.csv.
- Do not create AI interpretation before Layer 3 descriptive outputs exist.

9. Recommended Next Developer Role

Recommendation: The next active developer should be a Data Analyst Specialist. This role should focus on turning accepted Layer 2 outputs into descriptive tactical profiles, not machine-learning-first modeling.

- descriptive tactical profiles;
- tactical fingerprint schema design;
- stability analysis;
- team comparison;
- similarity scoring;
- explicit use of layer2_shape_metric_quality_flags.csv in all shape-metric analysis;
- optional clustering only later, after descriptive profiling is sound.

10. Recommended Next Technical Phase

Recommended next phase: Layer 3 v0.1 - Descriptive Tactical Profiling.

- define the first canonical tactical fingerprint schema;
- summarize team shape tendencies by phase;
- create interpretable match/team profiles;
- identify stable versus unstable tactical behaviors;
- compare teams using simple descriptive statistics;
- join and consume shape metric quality flags before using shape metrics;
- prepare structured outputs for future AI interpretation.

11. Final General Developer 1 Closeout Note

Closeout: The project has moved from raw dataset exploration to reusable ingestion and feature infrastructure. The final Layer 2 semantic bug was caught by football sanity validation, patched, revalidated, and protected with a non-destructive quality flag layer. Future work should build on the existing modules rather than replace them. The next development phase should preserve the same engineering discipline: validate first, document decisions, keep schemas controlled, consume quality flags, and avoid unsupported tactical claims.

12. Retirement Status

Decision: General Developer 1 can now retire cleanly after Lead Developer acceptance of this final v0.2.3 handoff. Layer 3 v0.1 descriptive tactical profiling can begin in a fresh developer chat, using the quality flag outputs as a required safety gate.